

The Standard Eigenvalue Problem to Model Coupled Pendula

Recall

$$\begin{array}{c} \text{mass matrix} \\ [M]_{2 \times 2} \end{array} [\ddot{u}(t)]_{2 \times 1} = - \begin{array}{c} \text{stiffness matrix} \\ [K]_{2 \times 2} \end{array} [\vec{u}(t)]_{2 \times 1}$$
$$\begin{array}{c} \downarrow \\ \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \end{array} \begin{bmatrix} \ddot{u}_1(t) \\ \ddot{u}_2(t) \end{bmatrix} = - \begin{array}{c} \downarrow \\ \begin{bmatrix} \frac{m_1 \cdot l}{g} & -k \\ -k & k + \frac{m_2 \cdot l}{g} \end{bmatrix} \end{array} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$$

$$m \cdot a(t) = \sum F$$

This is the classic form suggested by Newton's 2nd Law!

We factored out a negative so we could write:

$$[M]_{2 \times 2} [\ddot{u}(t)]_{2 \times 1} + [K]_{2 \times 2} [\vec{u}(t)]_{2 \times 1} = [\vec{0}]_{2 \times 1}$$

Let's start with the matrix form

$$M \ddot{u}(t) = -K \vec{u}(t)$$

To create a standard eigenvalue problem (SEP),
let's assume $m_1 = m_2$.

$$\Rightarrow \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{u}_1(t) \\ \ddot{u}_2(t) \end{bmatrix} = - \begin{bmatrix} \frac{m \cdot l}{g} & -k \\ -k & k + \frac{m \cdot l}{g} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$$

Since $m = m_1 = m_2 > 0$, we know M is diagonal
w/ positive main diagonal elements. Therefore,
 M is regular & invertible.

$$M \ddot{u}(t) = -K \vec{u}(t)$$

$$\Rightarrow \ddot{u}(t) = -M^{-1} K \vec{u}(t)$$

$$\Rightarrow \begin{bmatrix} \ddot{u}_1(t) \\ \ddot{u}_2(t) \end{bmatrix} = - \begin{bmatrix} \frac{1}{3l} & 0 \\ 0 & \frac{1}{3l} \end{bmatrix} \begin{bmatrix} \frac{m \cdot g}{2} + k & -k \\ -k & k + \frac{m \cdot g}{2} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$$

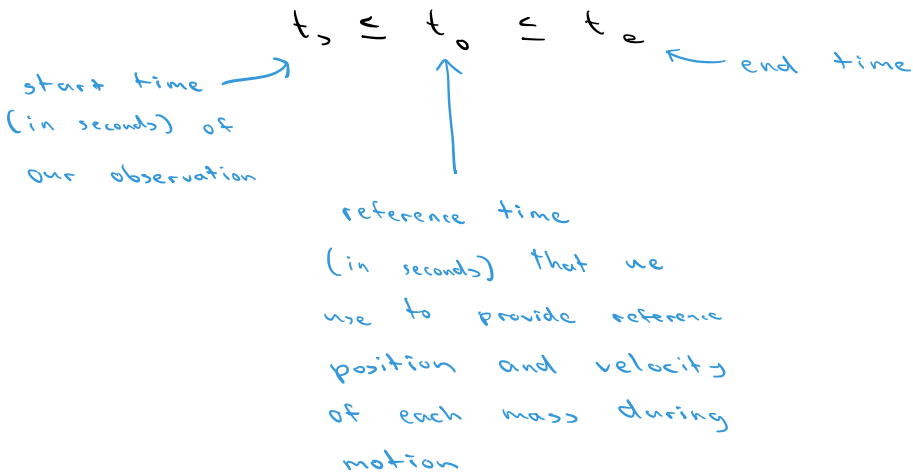
$$\Rightarrow \begin{bmatrix} \ddot{u}_1(t) \\ \ddot{u}_2(t) \end{bmatrix} = - \begin{bmatrix} \frac{g}{2} + \frac{1}{3l} & -\frac{1}{3l} \\ -\frac{1}{3l} & \frac{1}{3l} + \frac{g}{2} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$$

$$\Rightarrow \ddot{u}(t) = -A \vec{u}(t) \quad \text{where } A = M^{-1} \cdot K.$$

Suppose we watch our pendula move during some fixed time interval

$$t \in \underbrace{[t_s, t_e]}_{\text{start} \quad \text{end}}$$

Suppose we also want to choose a reference time in our interval $t_0 \in [t_s, t_e]$ where



Recall the goal of the problem is to find functions

$$u_1(t) = x_1(t) - x_1 \quad u_2(t) = x_2(t) - x_2$$

that predict the location of the center of mass at any time $t \in [t_s, t_e]$ during our observation.

Specifically, we determine the values of

$$\vec{u}(t_0) = \begin{bmatrix} u_1(t_0) \\ u_2(t_0) \end{bmatrix} = \begin{bmatrix} x_1(t_0) - x_1 \\ x_2(t_0) - x_2 \end{bmatrix} = \begin{bmatrix} x_1(t_0) \\ x_2(t_0) \end{bmatrix} - \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

reference displacement vector

reference position for masses

$$\Rightarrow \vec{u}(t_0) = \vec{x}(t_0) - \vec{x}$$

equilibrium positions

We also define the reference velocity vector:

$$\dot{u}(t_0) = \begin{bmatrix} \dot{u}_1(t_0) \\ \dot{u}_2(t_0) \end{bmatrix}$$

Assume that not all reference data is zero (masses are moving)

Looking back at

$$\ddot{u}(t) = -A u(t),$$

we ask: Do we know anything about the underlying system that might help us make a first guess about how to solve this problem?

- We know each mass is a bob on a pendulum operating in a small oscillation. In that phenomena, we saw that cosine functions arise.

What if we guessed that

$$\begin{aligned} u_1(t) &= v_1 \cdot \cos(\omega(t-t_0)) \\ &\quad \uparrow \quad \quad \quad \uparrow \\ &\quad \text{unknown amplitudes} \quad \text{reference time} \\ &\quad \text{of oscillation} \quad \quad \text{during motion} \\ u_2(t) &= v_2 \cdot \cos(\omega(t-t_0)) \end{aligned}$$

The scalars v_1, v_2 should be nonzero since we assumed masses are moving

where ω

$$\omega = \frac{2\pi}{T}$$

is an unknown "angular frequency" of oscillation with T being the period of one cycle of oscillation (measured in seconds).

What if we guessed that

$$\begin{aligned} \vec{u}(t) &= \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} = \begin{bmatrix} v_1 \cos(\omega(t-t_0)) \\ v_2 \cos(\omega(t-t_0)) \end{bmatrix} \\ &= \begin{bmatrix} \cos(\omega(t-t_0)) \cdot v_1 \\ \cos(\omega(t-t_0)) \cdot v_2 \end{bmatrix} \\ &= \cos(\omega(t-t_0)) \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \end{aligned}$$

$$\Rightarrow \vec{u}(t) = \cos(\omega(t-t_0)) \cdot \vec{v}$$

Here we assume $v_1, v_2 \in \mathbb{R}$ are constant scalars and thus $\vec{v} \in \mathbb{R}^2$.

This is called the

cosine
ansatz

$$\vec{u}(t) = \cos(\omega(t-t_0)) \cdot \vec{v}$$

Let's see what this would mean in our system

$$\ddot{\vec{u}}(t) = -A \vec{u}(t)$$

$$\Leftrightarrow \frac{d^2}{dt^2} [\vec{u}(t)] = -A \cdot \vec{u}(t)$$

$$\begin{bmatrix} \frac{d^2}{dt^2} u_1(t) \\ \frac{d^2}{dt^2} u_2(t) \end{bmatrix} = - \begin{bmatrix} \frac{g}{l} + \frac{k}{m} & -\frac{k}{m} \\ -\frac{k}{m} & \frac{k}{m} + \frac{g}{l} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix}$$

Let's analyze the left-hand side first using cosine ansatz.

$$u_i(t) = v_i \cdot \cos(\omega(t-t_0))$$

First Derivative

$$\begin{aligned} \frac{d}{dt} u_i(t) &= \frac{d}{dt} v_i \cdot \cos(\omega(t-t_0)) \\ &= v_i \cdot \frac{d}{dt} \cos(\omega(t-t_0)) \\ &= v_i \cdot -\sin(\omega(t-t_0)) \cdot \omega \\ &= -\omega \cdot v_i \cdot \sin(\omega(t-t_0)) \end{aligned}$$

Second Derivative

$$\begin{aligned} \frac{d^2}{dt^2} u_i(t) &= \frac{d}{dt} -\omega \cdot v_i \cdot \sin(\omega(t-t_0)) \\ &= -\omega \cdot v_i \cdot \frac{d}{dt} \sin(\omega(t-t_0)) \end{aligned}$$

$$= -\omega \cdot v_i \cdot \cos(\omega(t-t_0)) \cdot \omega$$

$$= -\omega^2 \cdot v_i \cdot \cos(\omega(t-t_0))$$

$$= -\omega^2 \cdot u_i(t)$$

Then, on the LHS we have

$$\ddot{u}(t) = \begin{bmatrix} \ddot{u}_1(t) \\ \ddot{u}_2(t) \end{bmatrix}$$

$$= \begin{bmatrix} -\omega^2 \cdot v_1 \cdot \cos(\omega(t-t_0)) \\ -\omega^2 \cdot v_2 \cdot \cos(\omega(t-t_0)) \end{bmatrix}$$

$$= -\omega^2 \cdot \begin{bmatrix} v_1 \cdot \cos(\omega(t-t_0)) \\ v_2 \cdot \cos(\omega(t-t_0)) \end{bmatrix}$$

$$= -\omega^2 \cdot \cos(\omega(t-t_0)) \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$= -\omega^2 \cdot \cos(\omega(t-t_0)) \cdot \vec{v}$$

Focusing on the RHS, we see

$$-A \vec{u}(t) = -A \cdot (\cos(\omega(t-t_0)) \cdot \vec{v})$$

$$= -\cos(\omega(t-t_0)) \cdot A \cdot \vec{v}$$

Using linearity

Then, we see if $u(t) = \cos(\omega(t-t_0)) \vec{v}$

$$\Rightarrow \ddot{u}(t) = -A u(t)$$

$$\Leftrightarrow -\omega^2 \cdot \cos(\omega(t-t_0)) \cdot \vec{v} = -\cos(\omega(t-t_0)) \cdot A \cdot \vec{v}$$

$$\Leftrightarrow \omega^2 \cdot \cos(\omega(t-t_0)) \cdot \vec{v} = \cos(\omega(t-t_0)) \cdot A \cdot \vec{v}$$

$$\Leftrightarrow \cos(\omega(t-t_0)) \cdot A \cdot \vec{v} - \cos(\omega(t-t_0)) \cdot \omega^2 \cdot \vec{v} = \vec{0}$$

$$\Leftrightarrow \cos(\omega(t-t_0)) \cdot (A \vec{v} - \omega^2 \cdot \vec{v}) = \vec{0}$$

We know that $\vec{v} \neq 0$ from our assumption that masses are moving.

The equation above must be true for all values of t .

But $\cos(\omega(t-t_0)) \neq 0 \in \mathbb{R}$ for all values of t .

$$\Rightarrow \ddot{u}(t) = -A \vec{u}(t)$$

$\Leftrightarrow$ We can find a nonzero vector $\vec{v} \in \mathbb{R}^2$ such that

$$A \vec{v} - \omega^2 \cdot \vec{v} = 0$$

$$\Leftrightarrow A \vec{v} = \omega^2 \cdot \vec{v}$$

$$\Leftrightarrow A \vec{v} = \lambda \vec{v} \text{ for } \lambda = \omega^2$$

This is a standard eigenvalue problem.

1h 40 m